



McMaster Research Ethics Board

Amendment

If the proposed amendment significantly alters the nature of the approved research, then the Chair may decide that a new application and REB review is required (See [TCPS2, Art. 6.16](#)).

If you are **ONLY** making administrative or other minor changes that do not substantively alter the currently approved protocol and documents, then you may be able to submit a **For Information Only** application to document the change, instead of an **Amendment**. The **For Information Only form** is only reviewed by the Ethics Secretariat staff to confirm the change does not require an **Amendment** form and ethics review. If you are unsure which form to use, please contact mreb@mcmaster.ca or ext. 23142.

Some examples of changes that may only require a **For Information Only** form include:

- Change to study title
- Change to investigator information
- Update of funding status
- Reporting clearance from other REBs
- Change to level of project
- Updating an open-ended interview guide (in certain cases, see [TCPS2, Art. 10.5](#))
- Minor or administrative changes to the Letter of Information or other study documents (e.g. new contact information, changing wording for clarity or to address typos/grammar, removing a survey question)

1.1 Date of Amendment Submission (for differentiating amendments in the Project Tree)

Jul/18/2026

1.2 Provide text that best describes this amendment, which will appear on the Project Tree. For example, Informed Consent, Recruitment, Data Security (maximum 100 characters)

Addition of Online Surveys

Sharing the Application

Before continuing, you may want to take a minute and **share** this application with the Principal Investigator, Faculty Supervisor, or whomever else will need access. Use the “**Share**” action button on the left side of the screen to give other researchers access and set their permissions. The person you are **sharing** the application with must have a MacID and already be registered in the MacREM system.

Summary of Revisions in Response to MREB Chair Comments

When you respond to the MREB ethics review sent by the MREB Chair, return to this question and select that option. This will allow you to upload your summary of revisions. If this is the initial submission of your application or you are submitting a revised version due to administrative comments from the Research Ethics Officer, then select one of those options instead.

1.3 Please select the status of your application:

- ☒ My revised amendment form in response to ethics review comments from the MREB Chair
- ☐ My initial amendment or my revised amendment form in response to the Research Ethics Officer administrative comments

1.4 Upload the document summarizing your revisions below (see #3 below). Only upload the summary document here, as revised supporting documents (e.g. letter of information) should be uploaded in the appropriate section of the form.

[96966]

		Documents					
Type	Document Name	File Name	Version	Date	Version	Size	View Delete
Response Documents	SummaryOfRevisionsAmendment	SummaryOfRevisionsAmendment.pdf	Jul/30/2026	2		271.4 KB	Download

1.5 Steps to respond to Chair comments.

MREB#: 8215 Title: State of the Practice for Scientific Software

1) Revise the MREB Form

- Refer to reviewer comments in the PDF sent via email by the Chair.
- Navigate to applicable questions within the online form and make the revisions.
- Edits to text fields in the main form should be made within the online interface so the system can track the changes (as opposed to making changes in a Word document and then pasting back into MacREM).

2) Revise Supporting Documents if Required (e.g. Letter of Information, Questionnaire, etc.)

- Use track changes so the Chair can easily see revisions.
- Upload a PDF version of the revised document(s), with a new date and version number, to the appropriate question within the application.
- Delete the old version of the document(s) from the application.

3) Summarize the Revisions for the Chair

- Create a document that lists each reviewer comment (and question number), along with your response to the comment. Follow the format in the Sample Summary of Revisions document (See Help/Templates in the top menu or click < [here](#) >).
- Use italics, or some other visible formatting, to distinguish your response from the reviewer comments.
- Any changes you make to the application that are in addition to responding to reviewer comments should be explained in the document as well.
- Upload the summary of revisions document above in question 1.5.

*** Once everything is done you can sign the form to resubmit to the Chair ***

Following the above steps will make it easier, and therefore faster, for the Chair to review the revisions. A more comprehensive guide to responding to reviewer comments using MacREM is available (See MacREM Quick Guide Respond to REB in Help/Templates or click < [here](#) >).

If you think it will be insufficient to use the summary document to address a concern you have with a specific reviewer comment, and that a discussion with the Chair may be necessary, please directly contact the reviewing Chair via email or phone.

Amendment Screening Questions

1.6

Is this amendment for an Instructor's Course-based application?

For making amendments to:

- Instructor or co-instructor contact information
- Course syllabus
- Student projects
- Sample recruitment, informed consent, test instruments

☐ Yes - Amendment to Instructor's Course-based application

☒ No - Amendment for Standard MREB Application

1.7

Are you making changes to any of the following General Information?

- Title
- Contact information or changes to investigators
- Funding status
- Timelines
- Location of research
- Other REB review or clearances
- Level of project

☐ Yes

☒ No

1.8

Are you making changes to the study design, including Project Description, Recruitment, Data Collection Methods, or Incentives?

- Project Description (Background and Purpose)
- Participants (characteristics or sample size)
- Conflict of Interest

MREB#: 8215 Title: State of the Practice for Scientific Software

- Research Involving Indigenous Peoples
- Recruitment
- Methods to Collect Data (e.g., data collection instruments)
- Secondary Use of Data
- Experience of researchers
- Incentives or compensation

☒ Yes

☐ No

1.9

Are you making changes to the Risks or Benefits involved in the research or the use of Deception/Partial Disclosure?

- Physical risks
- Psychological risks
- Social risks
- Community counseling or support services
- Deception/partial disclosure, debriefing the deception, and re-consent
- Benefits of the research

☐ Yes

☒ No

1.10

Are you making changes to the Consent Process, Participant Withdrawal, or How Study Results are Provided?

- Informed consent processes
- Assent process for individuals lacking capacity to consent
- Participant withdrawal from study
- Providing participants with study results

☒ Yes

☐ No

1.11

Are you making changes to the Confidentiality of Data or Data Security and Storage Procedures?

- Identifiability, security and storage details for administrative documents (participant contact information, signed consent forms, payment log, etc.)
- Identifiability, security and storage details for research data
- Transfer/Transport of study records
- Change to dissemination of findings or final disposition of study records

☐ Yes

☒ No

1.12 Provide a brief summary (max. 5 lines) of the changes you are making in this amendment:

I am adding an online survey option, in addition to the existing interview option, for study participants. We are having trouble recruiting interview participants because of the time commitment, but several potential participants offered that they would answer at least some of the questions if they had a survey option.

Changes to Existing Questions

Some questions on this page appear by default, even if you are not making changes to them. They are existing questions from the main application, most of which are editable for the purposes of this amendment. If you are making changes to these questions, go back to the screening question and check the box for General Information. If you are making an amendment to an Instructor's Course-based application, you can mostly ignore these questions.

Contact

2.4

Is this a post-doc, graduate or undergraduate student project? If so, the Faculty Supervisor's contact information will be requested and the Faculty Supervisor must sign the application prior to submission.

If the project supports both faculty and student research, then select NO, so the faculty researcher can be listed as the Principal Investigator.

☐ Yes

☒ No

If you are an Instructor for a course-based application, please answer NO to this question. For the purposes of this form, as an Instructor, you will be considered to be a Principal Investigator.

MREB#: 8215 **Title:** State of the Practice for Scientific Software

2.5 Principal Investigator

Title

First Name

Surname

Dr.

Spencer

Smith

Organisation

McMaster University

City

Hamilton

Telephone

9055259140 X27929

Email

smiths@mcmaster.ca

2.6 What is the faculty/department of the Principal Investigator?

Engineering - Computing

2.11 Are there any Co-Investigators?

☒ Yes

☐ No

2.12 Are there any Collaborators?

☐ Yes

☒ No

2.13 Are there any Research Assistants or Coordinators?

☐ Yes

☒ No

2.14 Co-Investigator

Title

First Name

Surname

Dr.

Jacques

Carette

Organisation

McMaster University

Department

Computing and Software

Faculty

Engineering

City

Hamilton

Telephone

9055259140

Email

carette@mcmaster.ca

2.14 Co-Investigator

Title

MREB#: 8215 **Title:** State of the Practice for Scientific Software

First Name

Surname

Dr.

Matthew

Giamou

Organisation

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Department

Computing and Software

Faculty

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City

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Telephone

905-525-9140

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giamoum@mcmaster.ca

2.14 Co-Investigator

Title

First Name

Surname

Dr.

Linyang

Chu

Organisation

McMaster

Department

Computing and Software

Faculty

Engineering

City

Hamilton

Telephone

905-525-9140

Email

chul9@mcmaster.ca

2.17 Are there any Student Investigators?

Do not include the Student Principal Investigator if listed in 2.3.

☒ Yes

☐ No

2.18 Student Investigator(s)

Title

First Name

Surname

Mr

Ziyang

Fang

Organisation

McMaster University

Department

Computing and Software

Faculty

Engineering

MREB#: 8215 **Title:** State of the Practice for Scientific Software

City
Hamilton

Telephone
905-525-9140

Email
fangz58@mcmaster.ca

2.18 Student Investigator(s)
Title
First Name
Surname
Mr.
Christopher
Schankula
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Computing and Software
Faculty
Engineering
City
Hamilton
Telephone
905-525-9140
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schankuc@mcmaster.ca

General Information

2.19 Provide the full title for your research project:
The State of the Practice for Research Software

2.19.1 Short Project Title

If your project title is long (more than 15 words), then consider providing a short title below. Otherwise, enter the full title of the project. The title entered below will appear on the project lists in MacREM and a shorter title can save "screen real estate". Please do not use an acronym for the short title.

State of the Practice for Scientific Software

2.20 The estimated review time for a new MREB application to receive ethics review feedback from the REB Chair is approximately 4 - 6 weeks. This does not include the initial administrative review to ensure the application is complete.

Keeping in mind the time for ethics review, enter the estimated date that you plan to begin recruiting participants. Or, for secondary use of data, enter the date you plan to receive the dataset.

Note that you cannot begin recruitment or data collection until you have received MREB ethics clearance ([TCPS2 Art. 6.11](#)).

Jun/18/2026

2.20.1 What is the estimated last date for data collection with human participants or, in the case of secondary data, when you intend to complete analysis for this project?

Dec/18/2026

Level of Project

2.21 For which Level of Project(s) will the data be used? (Check all that may apply)

☒ Faculty Research

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- ☐ Post-Doctoral Research
- ☐ Ph.D.
- ☒ Master's (Thesis, MRP)
- ☐ Staff/Administration Research
- ☐ Undergraduate (Thesis, Independent Study)
- ☐ Doctor of Practical Theology
- ☐ Other

Location

2.22 Select the location(s) where research will be conducted.

- ☒ McMaster University
- ☐ Community
- ☐ Hospital
- ☐ Outside Canada
- ☐ School Boards
- ☒ Online
- ☐ Other

Funding and Granting Agencies

2.23 Is this project currently being funded?

- ☐ Yes
- ☒ No

Project Description and Data Collection Procedures

3.1

Check the applicable box below if your amendment includes a change to that part of the study protocol. Check all that apply:

- ☐ Project description (background and purpose)
- ☒ Participants (characteristics or sample size)
- ☐ Conflict of Interest
- ☐ Research Involving Indigenous Peoples
- ☒ Recruitment
- ☒ Methods to collect data (e.g., data collection instruments)
- ☐ Secondary use of data
- ☐ Experience of researchers
- ☐ Incentives or reimbursement

Changes to the above may require revisions to the current study documents or the addition of new documents. For example, a change to the data collection methods would likely require revisions to the Letter of Information and potentially the recruitment documents.

3.3

For the change to participant details, please address the following.

- a) Describe the change and how it is different from what is currently approved.
- b) Explain the reason for the change.

In the original project our process was to invite one developer per scientific computing project. We would like to invite more potential subjects per project, since we are having trouble recruiting participants. Open source projects have multiple developers, not just one. Previously we looked for someone with 50 or more commits to the open source project. Now we would like to expand our pool to include more than one person, including individuals with between 10-49 commits. This should at least triple our number of potential contacts. The protocol will be the same for everyone contacted. Our original proposal said that we would contact potential participants twice. We would like to contact the original invitees a third time (except for the few who have already replied), providing them with the new option of an online survey (as discussed below). The new participants from the expanded pool will only be contacted twice, as in the original protocol, although the recruitment emails will be slightly modified (as shown in the revised documents).

3.6

For the change to the recruitment procedures, please address the following.

- a) Describe the change and how it is different from what is currently approved.
- b) Explain the reason for the change.

The contact will still be made via e-mail.

3.6.1 Do you have any new or revised documents related to the recruitment procedures?

- ☒ Yes
- ☐ No

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3.6.2 Upload any revised documents related to the recruitment procedures.

Upload in PDF format

The document file name should clearly indicate type of document, as the file name will appear on the clearance certificate (e.g., Student Recruitment Poster_6Feb2020_v1).

If uploading a revised document,

- 1) use track changes so the revisions can be easily verified
- 2) delete the previous version of the document from this upload question (if responding to a request for revisions)

[96967,96968,96970]

Documents							
Type	Document Name	File Name	Version Date	Version	Size	View	Delete
Amendment - Methods Participants Recruitment	RecruitingScript	RecruitingScript.pdf	Jul/30/2026	5	65.9 KB	Download	
Amendment - Methods Participants Recruitment	RecruitingScript_FollowUpEmail	RecruitingScript_FollowUpEmail.pdf	Jul/30/2026	5	67.7 KB	Download	
Amendment - Methods Participants Recruitment	RecruitingScript_ThirdFollowUpEmail	RecruitingScript_ThirdFollowUpEmail.pdf	Jul/30/2026	2	67.4 KB	Download	

3.7 For the change to the data collection methods, please address the following.

- a) Describe the change and how it is different from what is currently approved.
- b) Explain the reason for the change.

We are having trouble recruiting interview participants, probably in part because of the length of the interview. We would like to keep the interview option, but add an online survey option as well, for those that prefer this option. The questions in both will be the same, except the survey questions will be modified to use click-button options where possible to reduce the typing required by the participants. The survey is hosted on Microsoft forms through McMaster's Microsoft 365. The survey is available here: <https://forms.cloud.microsoft/pages/responsepage.aspx?id=B2M3RCm0rUKMJSjNSW9HcijtxID0eP9Et28dpBjT1etURFNCTFFWREhCMEo4NUJESVdKQ04zRUxGOS4u&route=shorturl> There is no survey question to collect the participant's name. However, the graduate students will know the set of people who were invited and the principle investigator will know what raw data is associated with which project. Although the identity of the participants are not made explicit, it would be possible to deduce a small set of possibilities from this available information. The LOI and consent preamble will be modified to make participants aware of this. The survey response data will be stored under Ziyang Fang's Microsoft 365 account. Only the two graduate students (Fang and Schankula) and the principle investigator (Smith) will have access to the data through their McMaster ID and authentication. At the conclusion of the retention period approved for this study, the researcher will delete the raw survey data by: 1. Deleting all responses from the Microsoft Form. 2. Deleting the associated Excel workbook containing the responses from OneDrive. 3. Removing the deleted items from the Microsoft 365 Recycle Bin and, if accessible, the Second-Stage Recycle Bin. Following these steps, the data will no longer be accessible to the research team through Microsoft 365. Residual copies may persist temporarily within McMaster University's managed Microsoft 365 backup and retention infrastructure in accordance with institutional policies, after which they are managed and disposed of by the institution and Microsoft according to their retention schedules. As for the interview transcript data, the raw survey data will be retained for 3 years after the study.

3.7.1 Do you have any new or revised documents related to the modified data collection methods (e.g. surveys, interview guides, experimental stimuli, Sonar educational debriefing sheet, etc.)?

☒ Yes

☐ No

3.7.2 Upload any revised documents related to the data collection methods.

Upload in PDF format

The document file name should clearly indicate type of document, as the file name will appear on the clearance certificate (Parent Interview Guide_6Feb2020_v1).

If uploading a revised document,

- 1) use track changes so the revisions can be easily verified
- 2) delete the previous version of the document from this upload question (if responding to a request for revisions)

[96971,96972]

Documents							
Type	Document Name	File Name	Version Date	Version	Size	View	Delete
Amendments - Test Instruments	SurveyConsentPreamble	SurveyConsentPreamble.pdf	Jul/30/2026	2	323.0 KB	Download	
Amendments - Test Instruments	Survey Questions	Survey Questions.pdf	Jul/30/2026	2	196.1 KB	Download	

3.11 After reviewing your proposed changes above, do you need to upload any additional revised or new documents that were not already uploaded elsewhere on the form (e.g., revised letter of information to account for changes to data collection methods)?

Note that if the document is already uploaded elsewhere in the amendment form, it should not be uploaded here.

- ☒ Yes
- ☐ No

3.12 Upload any new or revised documents related to the changes above, that are not already uploaded elsewhere on the form.

Upload in PDF format.

The document file name should clearly indicate type of document, as the file name will appear on the clearance certificate.

- If uploading a revised document,
- 1) use track changes so the revisions can be easily verified
- 2) delete the previous version of the document from this upload question (if responding to a request for revisions).

[96973,96974]

Type	Documents		Version Date	Version	Size	View	Delete
	Document Name	File Name					
Amendment - Methods Participants Recruitment	Oral-Consent-Script	Oral-Consent-Script.pdf	Jul/30/2026	5	93.3 KB	Download	
Amendment - Methods Participants Recruitment	ThankYouScript	ThankYouScript.pdf	Jul/30/2026	5	39.3 KB	Download	

Informed Consent Process

6.1 Check the applicable box below if your amendment includes a change to that part of the study protocol. Check all that apply:

- ☒ Informed consent process
- ☐ Assent process for individuals lacking capacity to consent
- ☐ Providing participants with study results
- ☐ Participant withdrawal from study
- ☐ Incidental findings/Third party disclosure

Changes to the above may require revisions to the current study documents or the addition of new documents. For example, a change to participant withdrawal details would likely require revisions to the Letter of Information.

6.2 For changes to the informed consent process, please address the following.

Changes to parental/surrogate consent and alternatives to obtaining consent can be addressed here as well.

- a) Describe the change and how it is different from what is currently approved.
- b) Explain the reason for the change.

In the original proposal the informed consent was verified via an oral script that is read before the interview begins. For those participants who select the survey option, the informed consent is the first question of the survey. If the user doesn't provide their consent, the survey ends. If they do provide consent, the survey continues. This change was needed because of the differences between an interview and a survey.

6.7 After reviewing your proposed changes above, do you need to upload any new or revised documents, that were not already uploaded elsewhere on the form (e.g. revised letter of information to account for changes to participant withdrawal details)?

- ☒ Yes
- ☐ No

6.7.1 Upload any new or revised documents related to the changes above, that are not already uploaded elsewhere on the form.

Upload in PDF format.

The document file name should clearly indicate type of document, as the file name will appear on the clearance certificate.

- If uploading a revised document,
- 1) use track changes so the revisions can be easily verified
- 2) delete the previous version of the document from this upload question (if responding to a request for revisions).

[96975,96976,96977]

Type	Document Name	File Name	Documents		Version Date	Version	Size	View	Delete
Amendment - Informed Consent	Letter-of-Information-Consent-Form2	Letter-of-Information-Consent-Form2.pdf	Jul/30/2026	5	94.5 KB	Download			

		Documents							
Type	Document Name	File Name	Version Date	Version	Size	View	Delete		
Amendment - Informed Consent	Survey-Consent-Script	Survey-Consent-Script.pdf	Jul/30/2026	2	109.5 KB	Download			
Amendment - Informed Consent	Oral-Consent-Script	Oral-Consent-Script.pdf	Jul/30/2026	5	93.3 KB	Download			

Principal Investigator Assurance

Principal Investigator Assurances

I understand that the following all constitute violations of the McMaster University’s Research Integrity Policy:

- failure to obtain research ethics clearance;
- carrying out research in a manner that was not cleared by one of the university’s REBs ([see TCPS2, Art. 6.11](#));
- failure to submit an **Amendment** to obtain ethics clearance prior to implementing substantive changes to a cleared study ([see TCPS2, Art. 6.16](#));
- failure to submit an **Annual Project Status form** in advance of the yearly anniversary of the original ethics clearance date; ([see TCPS2, Art. 6.14](#)).

Additionally, researchers are required to report **Adverse Events** (i.e., an unintended negative consequence or result affecting participants) to the MREB ethics office and the MREB Chair as soon as possible, and no more than 3 days after the event occurs ([see TCPS2, Art. 6.15](#)). A privacy breach affecting participant information should also be reported to the MREB ethics office and the MREB Chair as soon as possible. The Reportable Events form is used to document adverse events, privacy breaches, protocol deviations and participant complaints.

I confirm that I have read the [McMaster University Research Integrity Policy, and I agree to comply with this and other university policies, guidelines](#) and the Tri-Council Policy Statement (TCPS2) and the guidelines of my profession or discipline regarding the ethical conduct of research involving humans.

7.2 Signature of Principal Investigator

Request Signature

Sign

Signed: This form was signed by on

Signature Request: Signature requested from on

This form is enabled to auto-submit after all signatures have been obtained.

Before signing, please make sure that all uploaded documents are in PDF format. If you have a unique document type that will not properly convert to PDF, please contact the ethics office for assistance.

After signing please check to make sure the **green confirmation bar** appears before logging out or closing your browser. If you do not receive email confirmation of the submission within 24 hours, please notify us at mreb@mcmaster.ca.

If you send a signature request to someone and do not receive the email confirmation of submission within a few days, we recommend that you follow-up with the person and remind them to click on the link in the signature request email to review and sign the form.